

GPS TRACKING OF CARGO Made Easy For Logistics

ASHWINI KUMAR SINHA

We often require a central tracking system for delivering cargo and other large-quantity goods shipments in logistics. Real-time tracking and receiving notifications for shipments that are stuck or racing to their hub can be challenging. Tracking each shipment one by one on Google Maps and altering routes in case of obstructions or barriers can complicate matters further. Here is a system that integrates Google Sheets, providing real-time updates

for every parcel, logistics path, and current location. You can do real-time tracking of all the goods and easily share their tracking URL.

It is assumed that the logistics vehicle driver has a phone with Wi-Fi and internet connectivity for the device we have designed.

If you wish to use this device independently of a phone's Wi-Fi, you'll need to incorporate a GSM-based module (such as the SIM800L) into the device and make the necessary code modifications.

This allows the device to access the internet via GSM and upload data for cargo shipments at sea where internet access is limited. Alternatively, you can connect the device directly to satellite chips and adjust

BILL OF MATERIALS

Components	Quantity	Description
AIR 530 GPS	1	Serial GPS module
ESP chip 8285/8266	1	Microcontroller
Battery	1	3.3V Micro

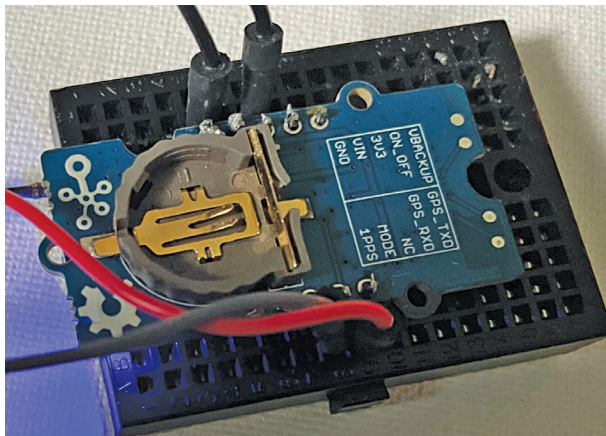


Fig. 1: GPS device attached with shipment

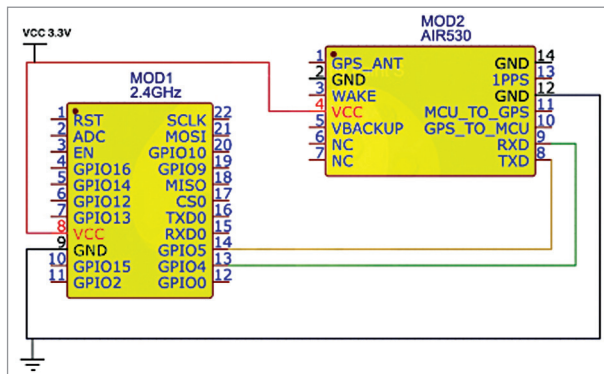


Fig. 2: Circuit diagram

```

1  function doGet(e) {
2  Logger.log( JSON.stringify(e) );
3  var result = 'Ok';
4  if (e.parameter == 'undefined') {
5  result = 'No Parameters';
6  }
7  else {
8  var sheet_id = '17FJp0kTV11ICqQMx-8hGBp1XArfXuG2PexziB1S8SjE'; // Spreadsheet ID
9  var sheet = SpreadsheetApp.openById(sheet_id).getActiveSheet();
10 var newRow = sheet.getLastRow() + 1;
11 var rowData = [];
12 var Curr_Date = new Date();
13 rowData[0] = Curr_Date; // Date in column A
14 var Curr_Time = Utilities.formatDate(Curr_Date, "Asia/", 'HH:mm:ss');
15 rowData[1] = Curr_Time; // Time in column B
16 for (var param in e.parameter) {
17 Logger.log('In for loop, param=' + param);
18 var value = stripQuotes(e.parameter[param]);
19 Logger.log(param + ':' + e.parameter[param]);
20 switch (param) {
21 case 'location':
22 rowData[2] = value; // Temperature in column C
23 result = 'location Written on column C';
24 break;
25 case 'destination':
26 rowData[3] = value; // Humidity in column D
27 result += ' destination Written on column D';
28 break;
29 case 'cargo-id':
30 rowData[4] = value; // Humidity in column D
31 result += ' cargo id Written on column D';
32 break;
33 default:
34 result = "unsupported parameter";
35 }
36 }
37 Logger.log(JSON.stringify(rowData));
38 var newRange = sheet.getRange(newRow, 1, 1, rowData.length);
39 newRange.setValues([rowData]);
40 }
41 }
42 return ContentService.createTextOutput(result);

```

Fig. 3: Deployment of the script as an app